

Lab Assignment-3

3. Programming Exercises on Classification Metrics and Models

- 3.1 Write a program to apply KNN Classification Model using Eigen, on the given data set to predict the class for a new datapoint P(4,3).

Data Point	x1	x2	y
P1	1	2	0
P2	2	3	0
P3	3	1	0
P4	6	5	1
P5	7	6	1

- 3.2 Write a program to create a confusion matrix from the given actual and predicted classes. Also, calculate accuracy, precision, recall, and F1 score using the confusion matrix.

Actual Class	Predicted Class
1	1
0	0
1	1
1	0
0	0
1	1
0	0
0	1
1	1
0	0